Class 39: Bose-Einstein Statistics

What can we say about the statistics displayed by particles that show quantum mechanical behavior, but *do not* obey the Pauli's exclusion principle? What are examples of such particles? Particles that have integer spin such as photons, several atoms, and even pairs of electrons under some conditions, fall under this category. The statistical behavior of these particles was described by Satyendra Nath Bose, and Albert Einstein, and goes by their names as Bose-Einstein statistics. Particles obeying these statistics are called Bosons.

Exemption from Pauli's exclusion principle implies that any number of particles can occupy the same state at the same time. In the derivation that follows, we are examining particles that are identical, indistinguishable, and exempt from the Pauli's exclusion principle.

Similar to the analysis in previous classes (Classes 12, 17, and 18) let us consider a system of particles that are allowed to occupy energy levels $\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4, ...\varepsilon_i$. Let us assume that there are s_i number of allowed states at ε_i, and that the total number of particles at ε_i is n_i. The values of s_i and ε_i are fixed for the system, since we have defined or identified a system with those values. The value of n_i is however variable, and it is our intention to find out what will be the equilibrium values of n_i, at each of the ε_i, given the various constraints we are placing on the system.

The possibility of placing any number of particles in a single state impacts our analysis of the system significantly. In fact, we can even have all of the particles in the same state at a given energy level. In other words we are free to arrange a collection of states and a collection of particles in any manner possible with almost no other restrictions – excepting one important restriction, the particles cannot exist independent of the states. What does this mean? Let us look at a more general example and see how we need to approach this situation mathematically. Assume we have 'X' objects of one kind and 'Y' objects of another kind. The number of different ways in which these can be arranged is given by:

$$\frac{(X+Y)!}{X!Y!}$$

The situation we are faced with is somewhat similar, but with one important difference. In the above example, we can have all of the 'X' objects followed by all of the 'Y' objects, as one of the arrangements. However in the case we are considering we are *excluding* the possibility that we can have an arrangement where we have all of the particles followed by all of the states, since that would imply that there are no particles in any of the states, and all the particles are outside all of the states. In the schematic in Figure 39.1 below, if the boxes represent states, and '*' represents particles that can occupy these states, we cannot have the situation shown below:

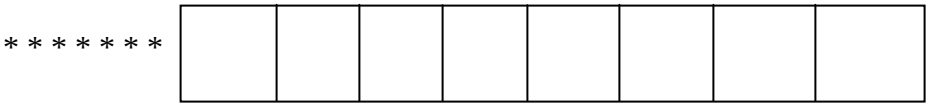


Figure 39.1: An arrangement of particles and states that is not permitted since it implies that all of the particles are outside all of the states

Therefore, the mathematics leads to acceptable results only if we conceptually treat the situation as one where we have n_i particles to be placed in (s_i-1) states. The implication being that if we have an arrangement where we have all of the n_i particles followed by all of the (s_i-1) states, all of the particles can be thought of as being in the first state. Similarly if we have an arrangement where we have all of the (s_i-1) states followed by all of the n_i particles, it would imply that all of the particles are in the last state. Alternately, if the s_i states are thought of as a large box with (s_i-1) partitions, similar to the schematic drawn above, all arrangements obtained from rearranging (s_i-1) partitions and n_i particles, are acceptable. The use of (s_i-1) partitions ensures that particles are never 'outside' all permissible states. The s_i states have (s_i-1) partitions between them.

Mathematically all of the acceptable conditions can therefore be expressed using:

$$\frac{(n_i + s_i - 1)!}{n_i!(s_i - 1)!}$$

The above expression is for a single energy level ε_i. if we denote this as $\Omega(n_i)$, then

$$\Omega(n_i) = \frac{(n_i + s_i - 1)!}{n_i!(s_i - 1)!}$$

Considering all of the possible energy levels, and the number of particles and states at these levels, the total number of microstates in which the system can exist while still at the same macrostate, is given by:

$$\omega(\overline{N}) = \Omega(n_1) \times \Omega(n_2) \times \Omega(n_3) \ldots = \prod_i \Omega(n_i) = \prod_i [\frac{(n_i + s_i - 1)!}{n_i!(s_i - 1)!}]$$

Since $n_i + s_i >> 1$, we can reasonably approximate the above to:

$$\omega(\overline{N}) = \prod_i [\frac{(n_i + s_i)!}{n_i!(s_i - 1)!}]$$

Our intent is to find the probability of occupancy of a state s_i, at the energy level ε_i, when the system is at equilibrium. This is given by the expression $\frac{n_i}{s_i}$. Based on the approach of statistical mechanics, the equilibrium state corresponds to the most probable state of the system. In particular, statistical mechanics assumes that the most probable state swamps the probability of all of the other states combined. Therefore, similar to the approach we have adopted in the earlier classes, we will proceed to determine the most probable state, which will correspond to the state where $\omega(\overline{N})$ is a maximum – i.e. it is the state which has the maximum possible

microstates. In the process of maximizing $\omega(\overline{N})$, we will obtain an expression for $\frac{n_i}{s_i}$ when $\omega(\overline{N})$ is a maximum. This is the answer we seek.

Also, as indicated earlier, since the function $\omega(\overline{N})$ and the function $\ln \omega(\overline{N})$, either both increase, or both decrease, their maxima occur at the same point $\overline{N}_o$. Therefore instead of maximizing $\omega(\overline{N})$, we can chose to maximize $\ln \omega(\overline{N})$, which is mathematically easier to handle, and arrive at the same desired result, $\overline{N}_o$.

$$\ln \omega(\overline{N}) = \sum_i \ln(n_i + s_i)! - \sum_i \ln n_i! - \sum_i \ln(s_i - 1)!$$

Using Stirling's approximation, which states that for large X

$$\ln X! = X \ln X - X$$

$$\ln \omega(\overline{N}) =$$
$$\sum_i \{(n_i + s_i)\ln(n_i + s_i) - (n_i + s_i) - (n_i \ln n_i - n_i) - [(s_i - 1)\ln(s_i - 1) - (s_i - 1)]\}$$

Maximizing $\ln \omega(\overline{N})$ implies, $\partial \ln \omega(\overline{N}) = 0$

Since s_i is a constant, and only n_i can be varied,

$$\partial \ln \omega(\overline{N}) = 0 = \sum_i [\ln(n_i + s_i) - \ln n_i]\partial n_i = 0$$

In addition, particle conservation, and energy conservation give us the following additional equations, respectively:

$$\sum_i \partial n_i = 0$$

$$\sum_i \varepsilon_i \partial n_i = 0$$

Applying the Lagrange method of undetermined multipliers, discussed in greater detail in Class 12, we obtain:

$$-\ln(\frac{s_i + n_i}{n_i}) + \alpha + \beta\varepsilon_i = 0$$

Note: The negative sign in the first term in the equation above, is inserted by convention, to make the final results easier to interpret. Since the equations leading up to the above are all equal to zero, introduction of a negative sign before summing up is acceptable.

Simplifying further,

$$\frac{s_i}{n_i} + 1 = e^{\alpha + \beta\varepsilon_i}$$

Therefore:

$$\frac{n_i}{s_i} = \frac{1}{e^{\alpha + \beta\varepsilon_i} - 1} = f_{BE}$$

The Bose-Einstein distribution predicts that at very low temperatures, very nearly absolute zero, all particles in the system will condense into a single state, called the Bose-Einstein condensate. This was a prediction made around the year 1924. At that time, the experimental facilities available were not capable of attaining the low temperatures required to test this prediction. While it is not very difficult to attain temperatures of the order of 4 K using liquid He, getting to lower temperatures is a significant experimental challenge. It took almost 70 years from the time of the prediction, for experimental facilities to evolve to the stage where it was possible to demonstrate the formation of this condensate. In 1995 Eric Cornell and Carl Weiman, demonstrated the formation of the condensate using a small collection of Rb atoms, which they cooled to a few hundred nano Kelvin. For demonstrating the formation of the Bose-Einstein condensate, they were awarded the Nobel prize in Physics for the year 2001 along with Wolfgang Ketterle.

Incidentally, the mean background temperature of the universe is of the order of 2.7 K. Therefore temperatures below this are not expected to occur naturally anywhere in the Universe. Seen from that perspective, to attain an experimental condition outside the natural ability of the Universe, is itself a remarkable accomplishment.

The prediction of the Bose-Einstein condensate, is an example of the power of theoretical studies. Surprising predictions can be made of unknown states of matter and it could take years to decades to test the prediction.